



KAUSHAL VIVEK ABOJWAR

B.E. - Instrumentation & Control Engineering

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Pune, Maharashtra, India - 411033



EDUCATION

D.Y. Patil College of Engineering, Akurdi

2023 - 2027

B.E. - Instrumentation & Control Engineering | CGPA: 8.32 / 10

Shree Maharishi Vidya Mandir, Chandrapur, Chandrapur

2023

12th | CBSE | Percentage: 79 / 100

Shree Maharishi Vidya Mandir, Chandrapur, Chandrapur

2021

10th | CBSE | Percentage: 91.60 / 100

INTERNSHIPS

ShreeTech Data Limited | Electrical / Switchgear

06 Jan, 2026 - 06 Feb, 2026

Engineer

Learned an control sytem and panels

PROJECTS

Developed an Arduino-based irrigation system using real-time soil-moisture sensing for efficiency

Mentor: Mrs. Utkarsha Kharade | Team Size: 4

Achieved ~18% water savings, reduced irrigation cycle time (3.23 ± 3.02 min) by threshold-based pump control
Automated irrigation using moisture-sensor feedback (~669–680) to maintain soil moisture and control the pump
Validated performance with higher discharge rates (7.67 & 7.42 ml/s) using Arduino-UNO, FC-28, and 12V pump

Developed an AI-enabled IoT system that uses sensor data and ML to detect food spoilage in real time

Mentor: Guide: Dr. B.B. Musmade | Team Size: 5

Integrated MQ135, MQ3, MQ4 and DHT22 sensors to capture real-time gases & climate data for spoilage detection
Developed an OpenCV & TensorFlow Lite module to detect visual spoilage cues and use pH data for classification
Implemented an ESP32 IoT pipeline streaming sensor data to Blynk/Firebase for real-time spoilage prediction

Advanced Railway Infrastructure Monitoring System Leveraging IoT Sensors and Machine Learning Models

Mentor: Nitin Mane | Team Size: 1

Engineered an IoT system capturing vibration, temperature, and pressure data via LPWAN and Wi-Fi networks
Processed multi-sensor data using SVM and RNN models for early fault detection (~85–90% accuracy)
Enabled predictive maintenance, reducing unplanned downtime (~20–25%) and improving operational safety

PLC-Based Predictive Maintenance System with HMI Dashboard

Mentor: Dr. Yogesh Patil | Team Size: 3

Key Skills:

PLC Programming Structured Text mapp view B&R Automation Studio HMI SCADA Industrial Automation
Industrial Control Systems State Machine Design Role-Based Access Control Motor Control Systems
Alarm Management Systems

Designed and developed a predictive maintenance system for industrial motor control, built on B&R Automation Studio with the mapp View HMI framework, selected as a competing project in B&R's Open Automation Challenge 2026.

Unplanned motor failures in continuous manufacturing (e.g. conveyor and robotic arm systems in automotive plants) can halt production lines at a cost of ₹10–15 lakhs per hour. Traditional reactive alarm systems only respond after failure. This project shifts that model to a predictive one, inspired by inefficiencies I observed firsthand during an industrial internship.

IMs

- Other - <https://kaushalvabojwar-gif.github.io/Portfolio1/>

PERSONAL DETAILS

Gender: Male

Marital Status: Single

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Date of Birth: 04 Dec, 2004

Known Languages: English, Marathi, Hindi

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